

AP CSP | Lesson 13: Counted Loops

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Trace statements in their execution order. Each program statement and each brace occupies its own line. Show reasoning and use the stated valid input range.

1. x begins at 4. After $x \leftarrow x + 2$, what is its value? [1 mark]

A. 2 B. 4 C. 6 D. 8

Answer: _____

2. What does REPEAT 4 TIMES do to the statements inside its braces? [1 mark]

A. Runs the complete body four times in sequence. B. Runs only the first statement four times.
C. Executes the body until four outputs have appeared. D. Runs the body once and multiplies all results by four.

Answer: _____

3. What is displayed? [1 mark]

```
count ← 0
REPEAT 3 TIMES
{
  count ← count + 2
}
DISPLAY(count)
```

A. 2 B. 3 C. 5 D. 6

Answer: _____

4. A loop should add three inputs into total. Where should $\text{total} \leftarrow 0$ go? [1 mark]

A. After the final DISPLAY only. B. Before the loop, once.
C. Immediately before every addition inside the loop. D. It is never needed before $\text{total} \leftarrow \text{total} + \text{value}$.

Answer: _____

5. Which TWO statements are correct? Select exactly TWO. [1 mark]

A. REPEAT 0 TIMES skips the body. B. A fixed-count loop always performs at least one iteration.
C. Two outer iterations with three inner iterations each perform six inner-body executions. D. Two outer iterations with three inner iterations each perform five inner-body executions.

Answer: _____

6. count starts at 1. A loop repeats three times; each iteration first displays count, then adds 2 to count. What are the outputs in order? [1 mark]

A. 3, 5, 7 B. 1, 1, 1 C. 1, 3, 5 D. 1, 3, 5, 7

Answer: _____

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Written responses | Questions 7-8

7.

[4 marks]

Trace the program. Give its three outputs in order and the final value of count.

count $\leftarrow$ 2

REPEAT 3 TIMES

{

 DISPLAY(count)

 count $\leftarrow$ count + 3

}

8.

[4 marks]

The input queue contains 8, then 12, then 5. Give total after each iteration and the final displayed value.

total $\leftarrow$ 0

REPEAT 3 TIMES

{

 value $\leftarrow$ INPUT()

 total $\leftarrow$ total + value

}

DISPLAY(total)

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Written responses | Questions 9-10

9.

[4 marks]

A program sets $\text{count} \leftarrow 1$, then repeats $\text{count} \leftarrow \text{count} + 1$ four times. The intended final count is the number of completed iterations.

- (a) State the actual final count.
- (b) State how many iterations execute.
- (c) Correct the initialization.
- (d) Explain why placing $\text{count} \leftarrow 0$ inside each iteration would not be a correct repair.

10.

[4 marks]

An outer loop repeats twice. Its body first runs an inner loop three times, displaying "Tick" each time, then displays "Done" once after that inner loop.

- (a) How many Tick outputs occur?
- (b) How many Done outputs occur?
- (c) Write the complete output order.
- (d) Explain why Done is not displayed after every Tick.

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Written responses | Questions 11-12

11. [4 marks]

Initially *i* is 1 and total is 0. Repeat three times: add *i* to total, then increase *i* by 1.

(a) Give final total and final *i*.

(b) If the two body statements are swapped, give final total and explain which values are now added.

12. [4 marks]

Write AP pseudocode that reads two valid whole-number minute values from 0 through 180 using REPEAT 2 TIMES, calculates their average, and displays the average once. Give the expected output for inputs 30 and 50, and for inputs 0 and 0. Each brace occupies its own line.
